Computer Programming II (CSE2035)
Lab #8
Template

Prof. Jaeseung Choi
Dept. of Computer Science and Engineering
Sogang University

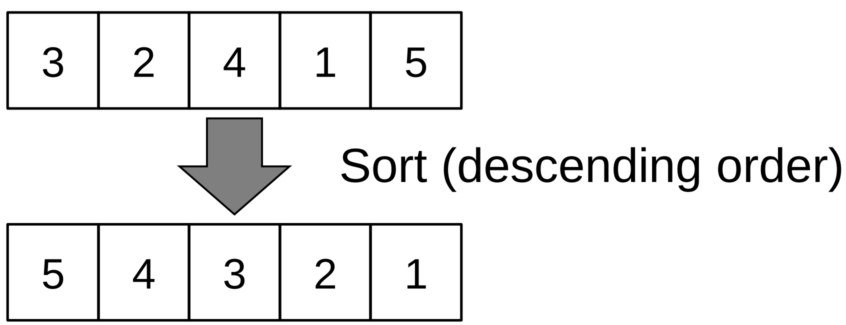
Homework 8-1: Sorting (10pt)

- **First, let's implement a function template `find_max()` in `max.hpp` file**
  - Argument "n" is the length of the array argument "`arr`"
  - Find the maximum element from "`T arr[]`" and swap that element with "`arr[0]`"
  - Other elements must remain unchanged
  - If "`arr`" is empty (and "n" is 0), don't have to do anything

```
template <typename T>
void find_max(T arr[], int n) {
    ...
}
```

Homework 8-1: Sorting (10pt)

- **Next, we will implement a simple sorting algorithm (known as *"selection sort"*) by using `find_max()`**
- **Here is the idea:**
  - (1) Find the maximum element from `arr[0]` to `arr[n-1]` and swap it with `arr[0]`.
  - (2) Find the maximum element from `arr[1]` to `arr[n-1]` and swap it with `arr[1]`.
  - (3) … (continue until we reach the end of `arr[]`)



Homework 8-1: Sorting (10pt)

- **So now let's implement another function template `simple_sort()` in `sort`.hpp file**
  - Argument "n" is the length of the array argument "`arr`"
  - Sort the elements in "`T arr[]`" in a descending order
  - If "`arr`“ is empty (and "n" is 0), don't have to do anything
- **This can be done easily by reusing `find_max()`**
  - But it's not a requirement

```
template <typename T>
void simple_sort(T arr[], int n) {
    ...
}
```

Homework 8-2: DynamicArray (10pt)

- **Do you remember the `MyString` class that we implemented in Lab #5?**
- **This time, we will implement a class template to create dynamic array that can store different types of data**

```
template <typename T>
class DynamicArray {
private:
    T *ptr;
    int length;
public:
    ...
};
```

Homework 8-2: DynamicArray (10pt)

- Properties, constructor, and `Print()` are given in `dynamic-array`.hpp file
- Do not change these properties and functions

```
class DynamicArray {
    ...
public:
    DynamicArray(T arr[], int len) { ... }
    void Print() { ... }
    ...
};
```

Homework 8-2: DynamicArray (10pt)

- **Implement the remaining methods in the same file (`dynamic-array.hpp`)**
  - Detailed specifications are given in the comment, so please read them carefully
  - Make sure that there is no memory leak, as we did for `MyString` class (you will lose point if memory leak occurs)

```
public:
    ...
    ~DynamicArray();
    void Append(DynamicArray *da);
    DynamicArray* Subseq(int idx, int n);
};
```

Submission Guideline

- **Submit the completed `max.hpp`, `sort.hpp` and `dynamic-array.hpp` files to "Lab #8 Assignment" in cyber campus**
  - You can fix `main.cpp` to test your code, but don't submit it
  - Do not zip the directory, just upload these files directly
  - **Please do not change the name of files**
  - If the submitted file doesn't compile with the "`make`" command, cannot give you any point
- **Deadline is 11/18 (Fri) 23:59**
- **Late submission is allowed until 11/21 (Mon) 23:59**
  - 20% deduction
- **Resubmission will be allowed until 11/25 (Fri) 23:59**
  - 40% deduction

Q & A Session